ID & Name: 150720062 – Mehmet Ali KAMALAK

Pre-LAB Work

1. For the circuits given in Fig. 1, calculate the low (f_L) and high (f_H) cut-off frequencies. Assume β=200, Cμ=3pF, Cπ=15pF. (Vcc=15V and Vee=-15V)

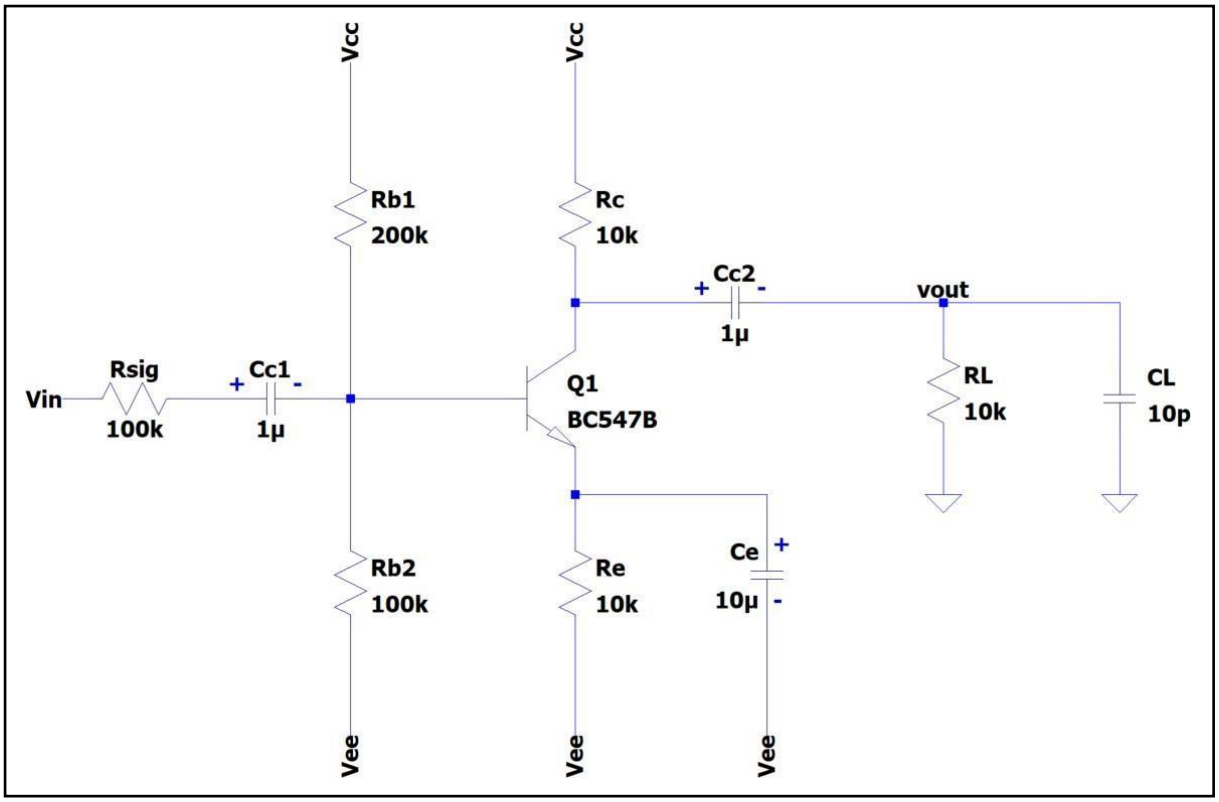
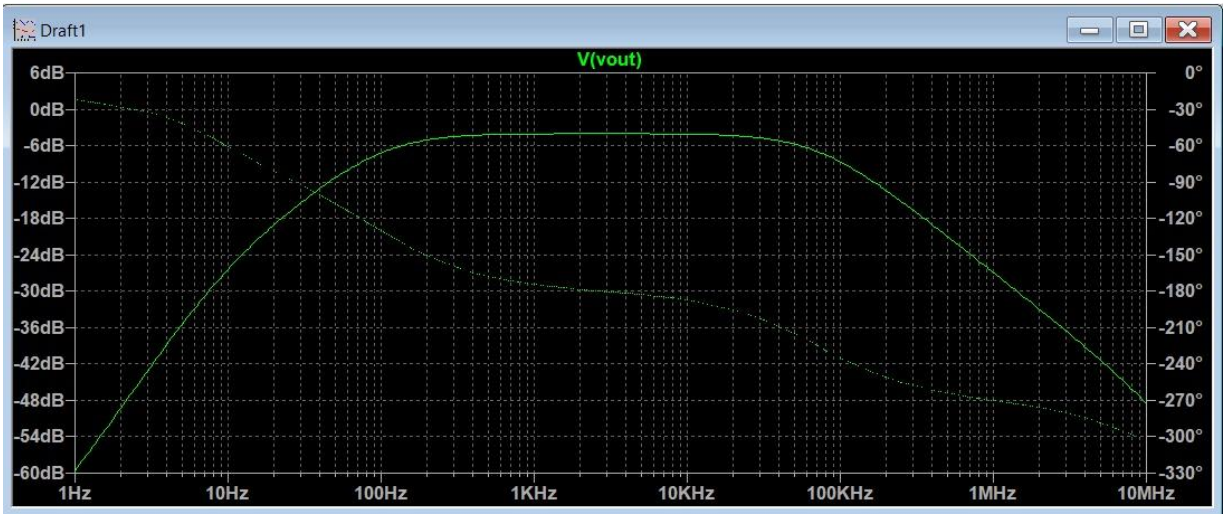


Figure 1

Answer: F low = 105,7 (Hz) F high = 88K (Hz)

2. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Conduct an AC analysis using the following spice command and plot the amplitude of the frequency response. Indicate the low (f_L) and high (f_H) cut-off frequencies.

   .ac dec 10 1 10e6 F low = 103(Hz) F high = 72K (Hz)



3. For the circuits given in Fig. 2, calculate the low (f_L) and high (f_H) cut-off frequencies. Assume β=200, Cμ=3pF, Cπ=15pF. (Vcc=15V and Vee=-15V)

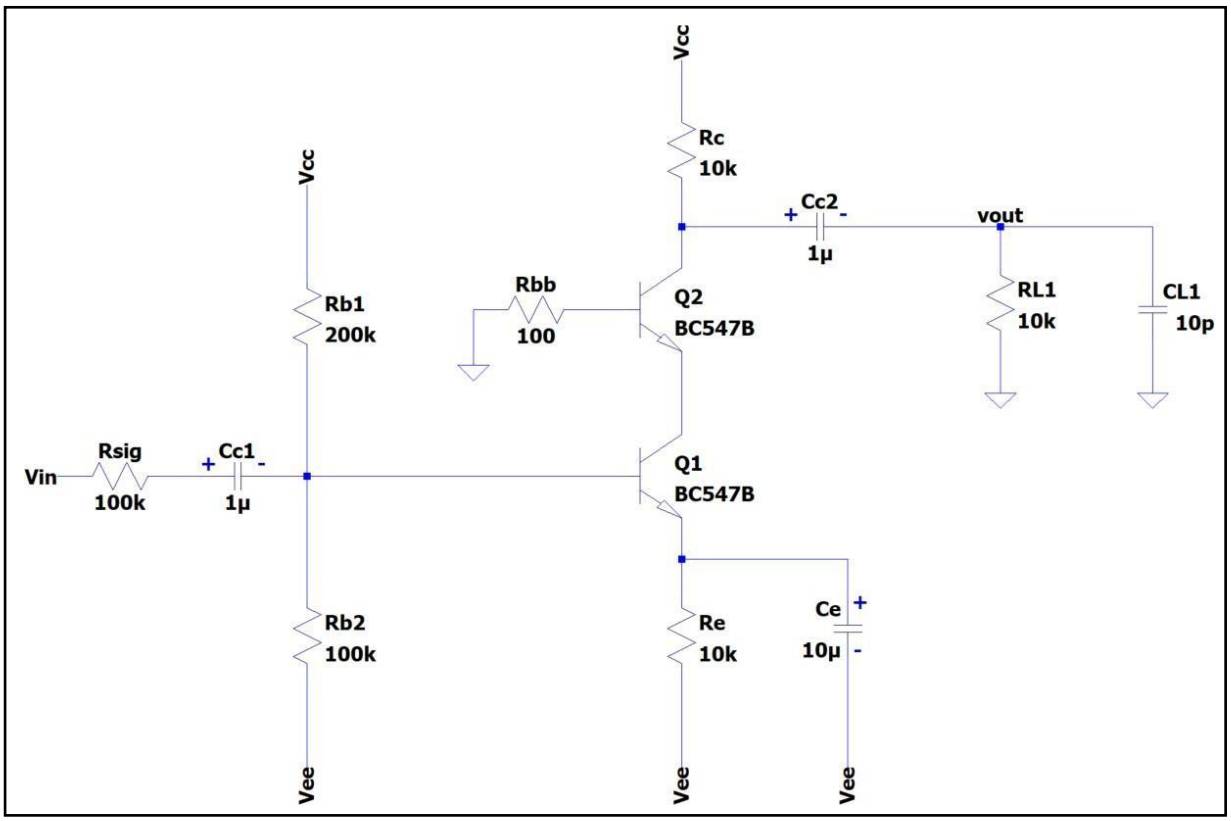
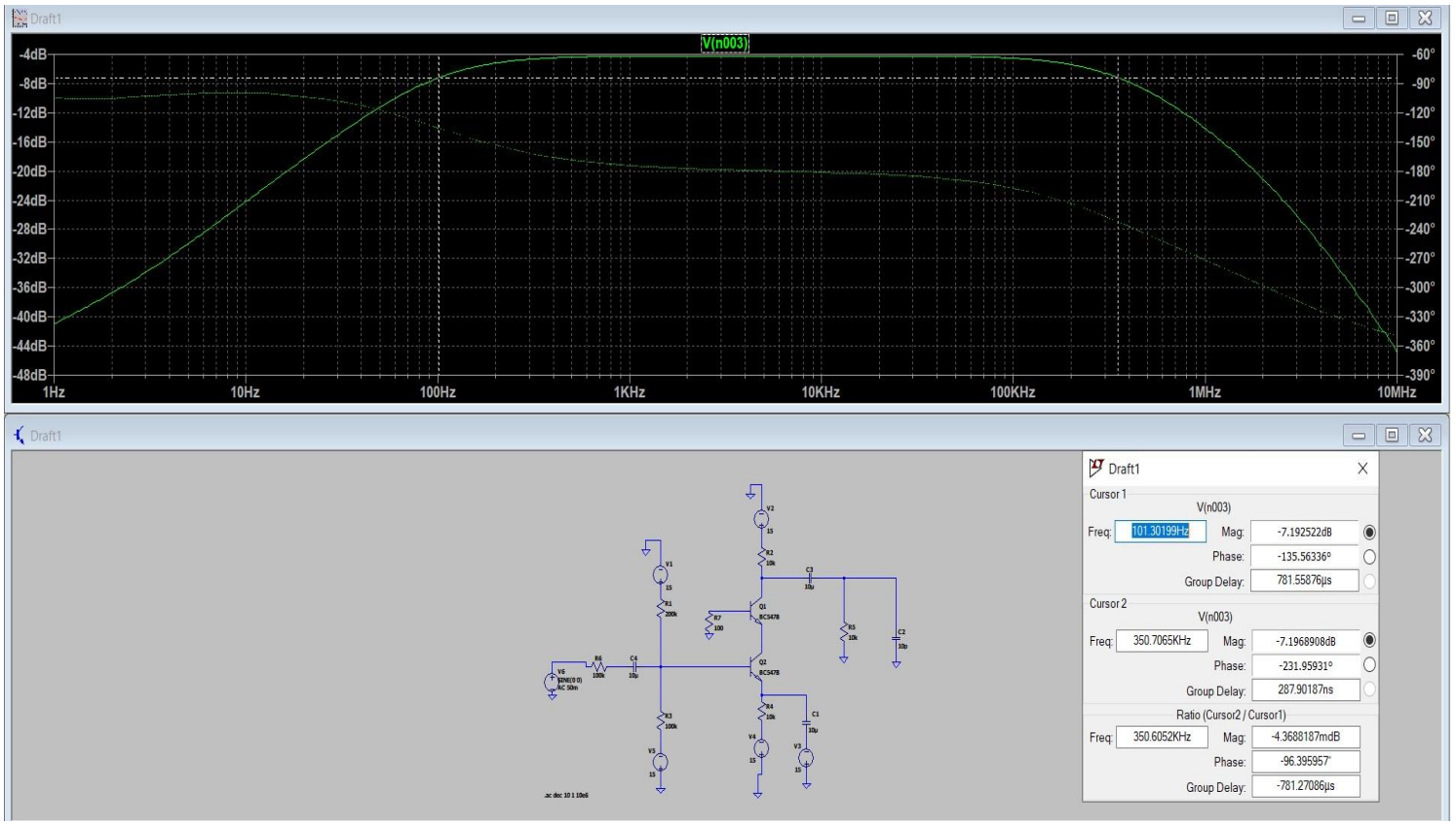


Figure 2

Answer: F low = 100 (Hz) F high = 340K (Hz)

4. Simulate the circuit in Fig. 2 with SPICE. Let Vcc=15V and Vee=-15V. Conduct an AC analysis and plot the amplitude of the frequency response. Indicate the low (f_L) and high (f_H) cut-off frequencies. F low = 101(Hz) F high = 350K (Hz)



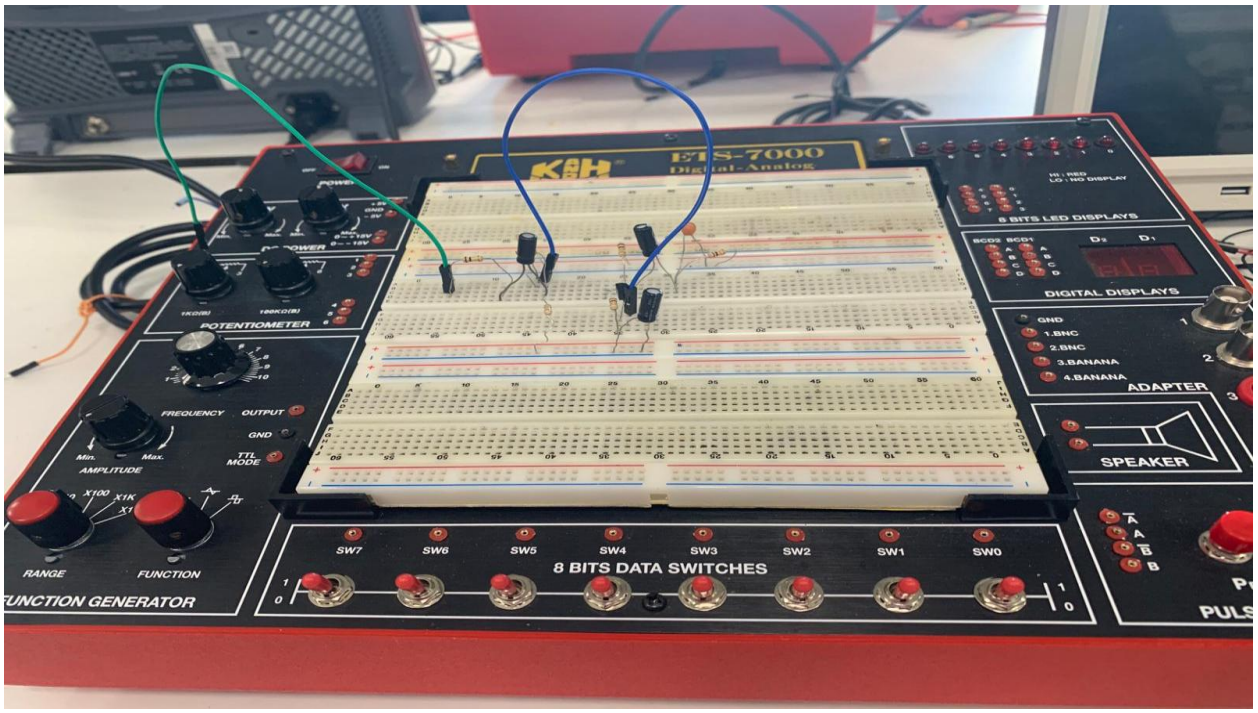
Equipment/Component List

Table 1: Equipment/Component List

Equipment/Component	Model/Value	Count
Breadboard	-	1
Resistor	100Ω	1
Resistor	10kΩ	**3**
Resistor	100kΩ	**2**
Resistor	200kΩ	1
Capacitor	10pF	1
Capacitor	1µF	2
Capacitor	10µF	1
BJT (npn)	BC547	**2**
Multi-meter	-	1
Function Generator	-	1
Oscilloscope	-	1
Voltage Source	15V, -15V	1
Probes & Cables	-	-

Experiment Procedure

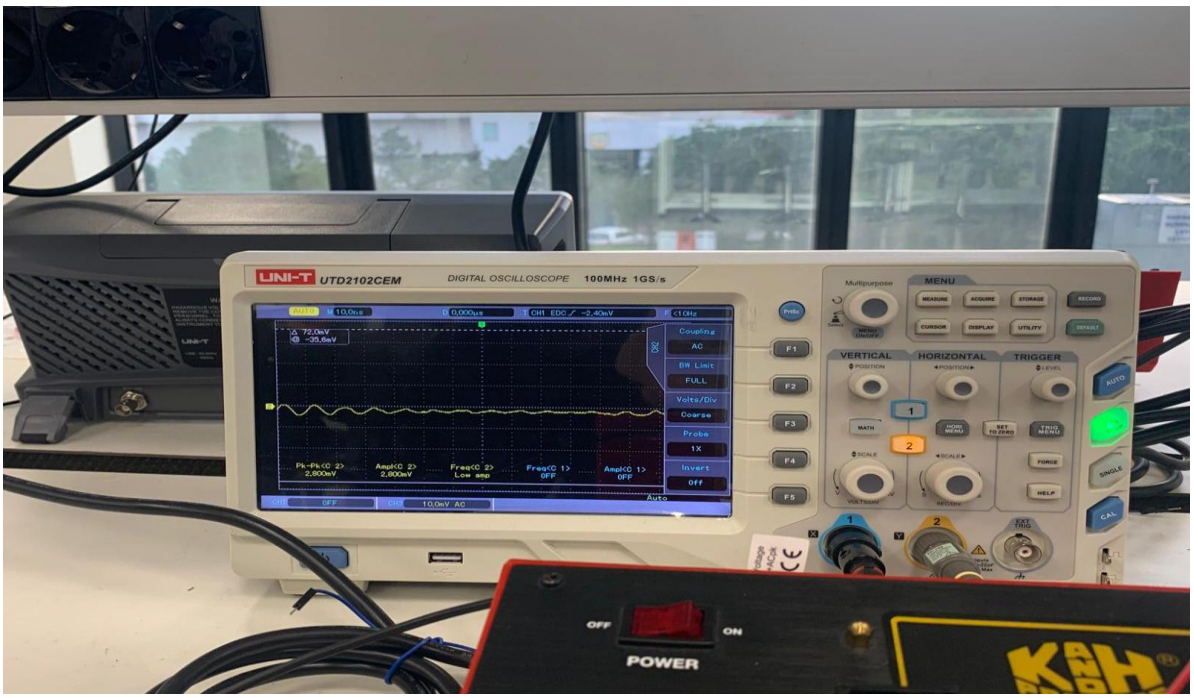
5. Adjust Vcc to 15V and Vee to -15V. You will also need the ground connection. Do not connect the supplies.
6. Set-up the circuit in Fig. 1. Be very careful while connecting the capacitor. The +/- pins must be connected as shown in the Fig. 1.
7. Connect the ground and then Vee and Vcc.
8. Apply sine waves of 10Hz, 100Hz, 1kHz, 10kHz, 100kHz, 1MHz, 10MHz with amplitudes of 50mV to Vin.
9. Also find the 3-dB cut-off frequencies.
10. Plot the gain in dB as a function of the frequency. Indicate the low (f_L) and high (f_H) cut-off frequencies. **F low ~ 300 Hz** **F high ~10.1K (Hz)**



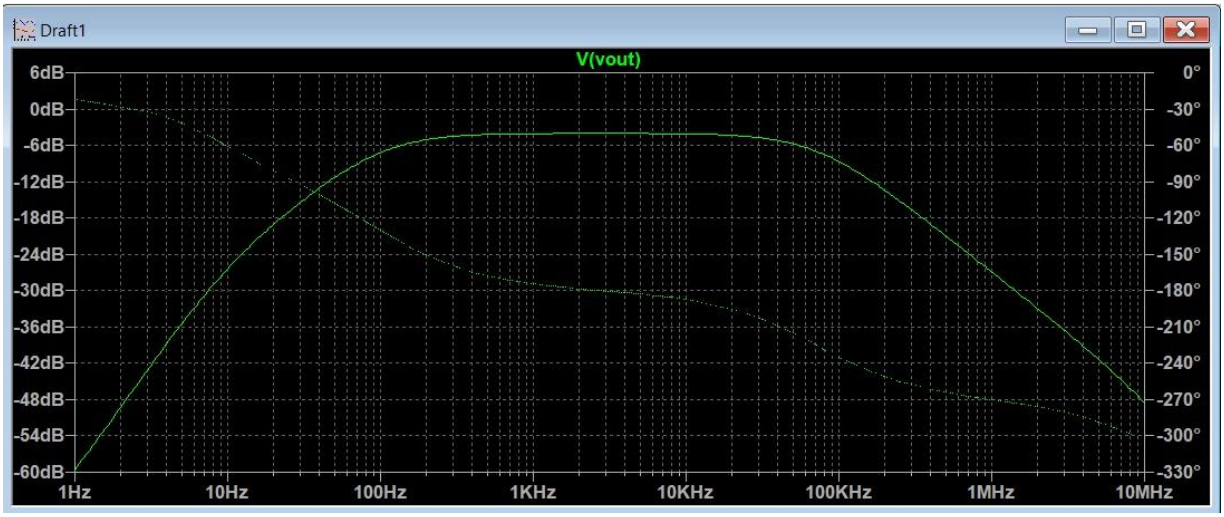
Lab and LTSpice Frequency Response Results

1.Devre

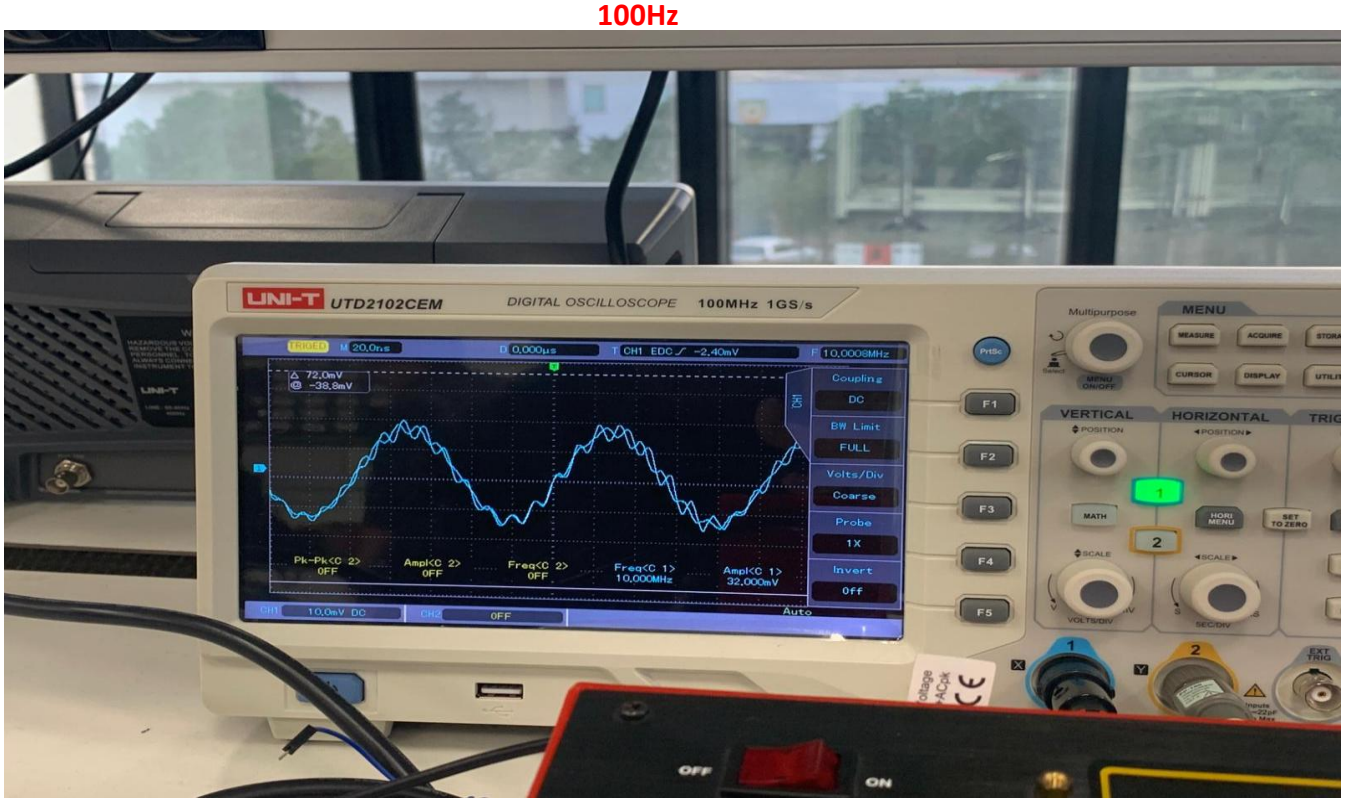
10HZ

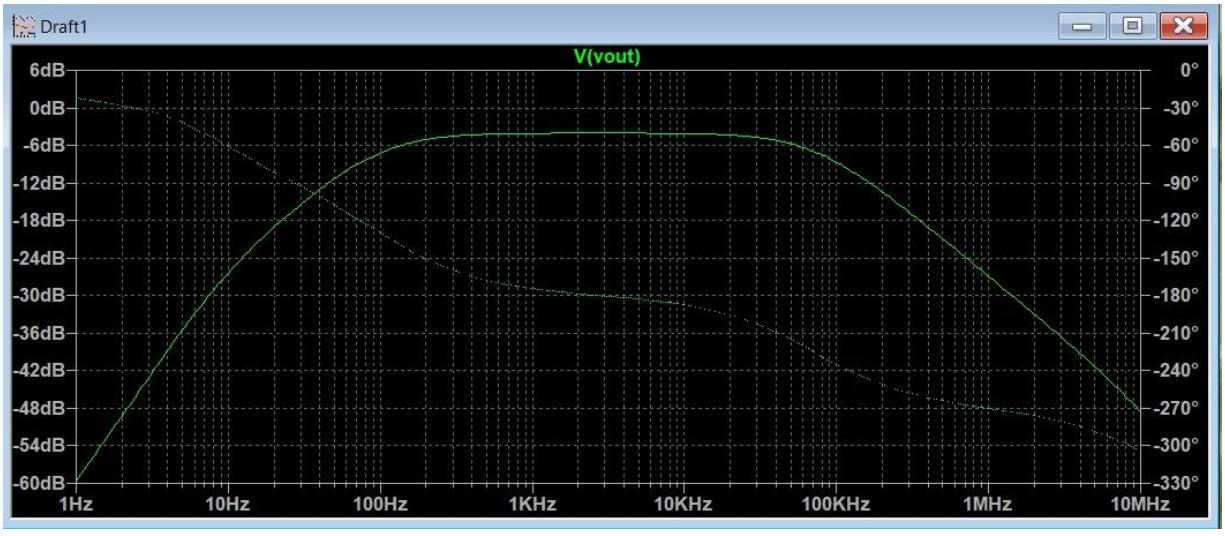


Freq. Response of Amplifiers Experiment 3



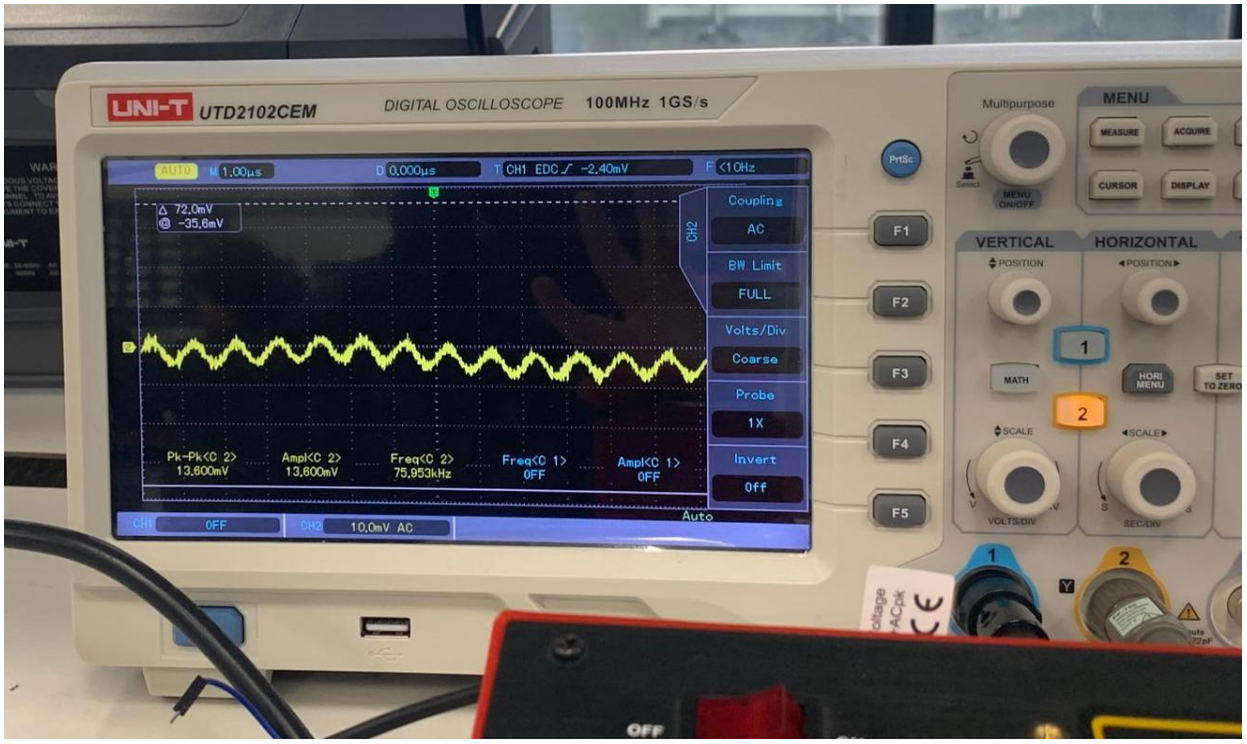
100Hz

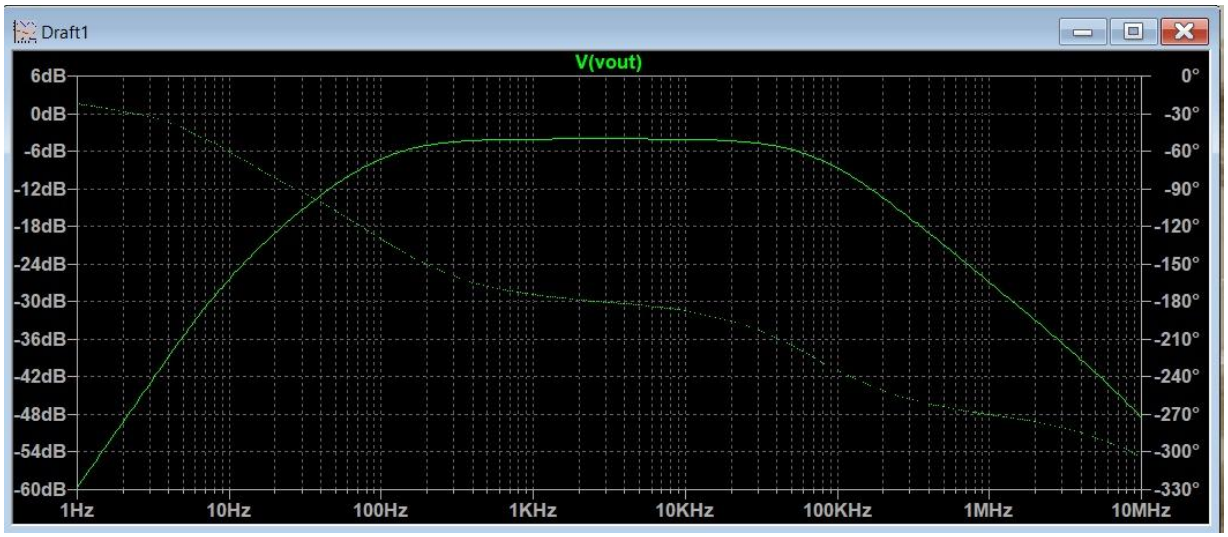




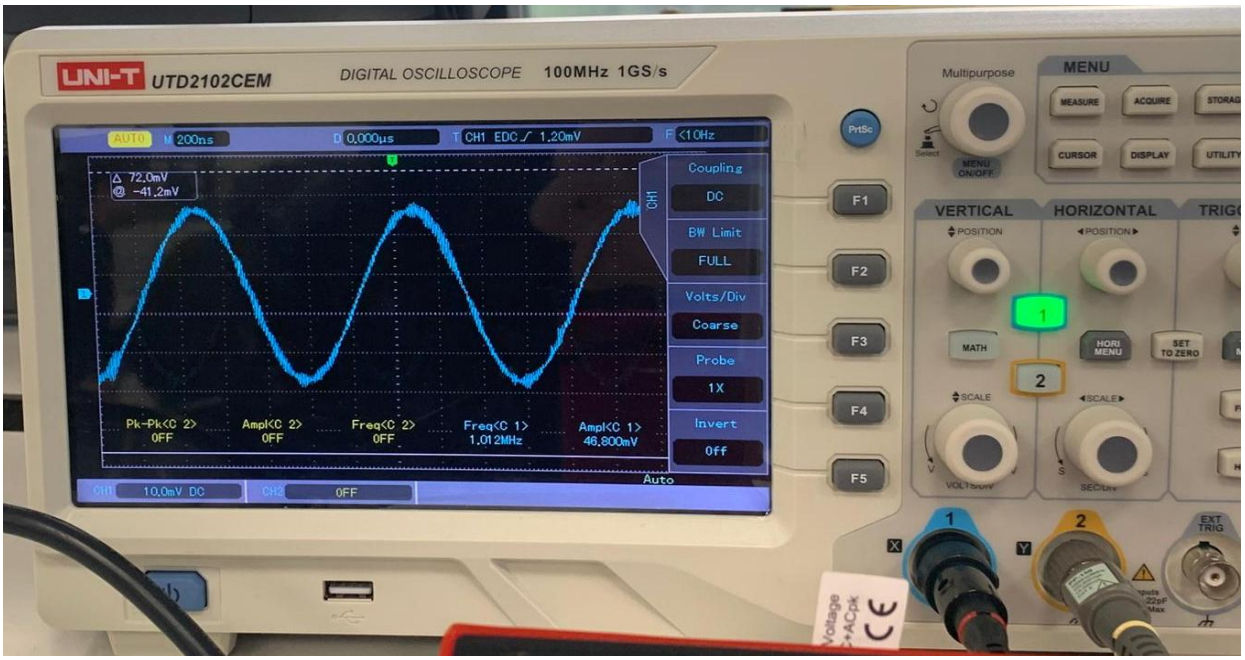
Freq. Response of Amplifiers Experiment 3

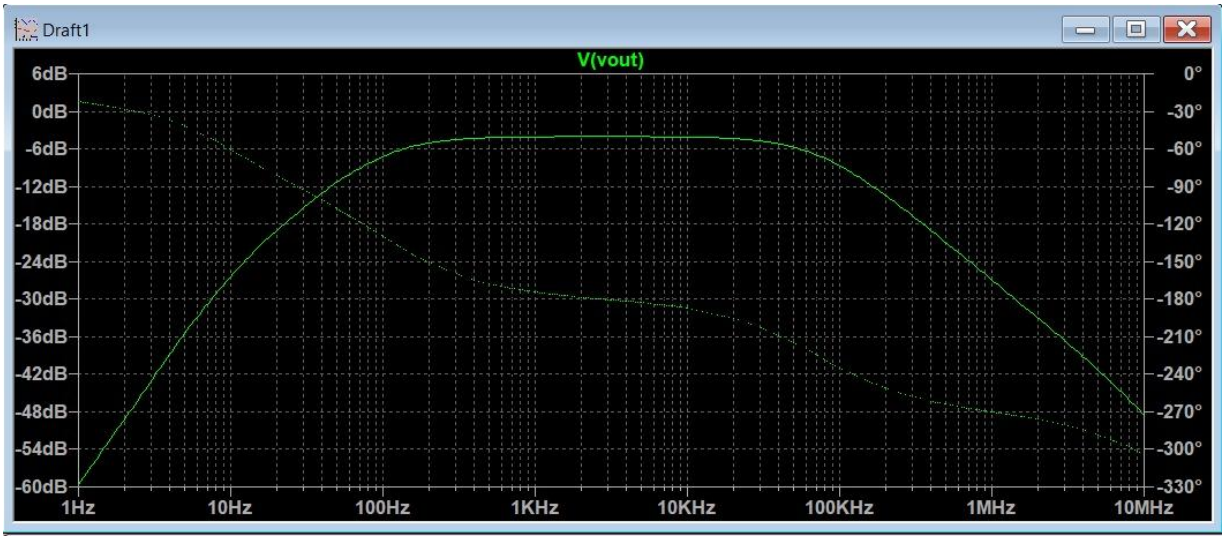
1k Hz



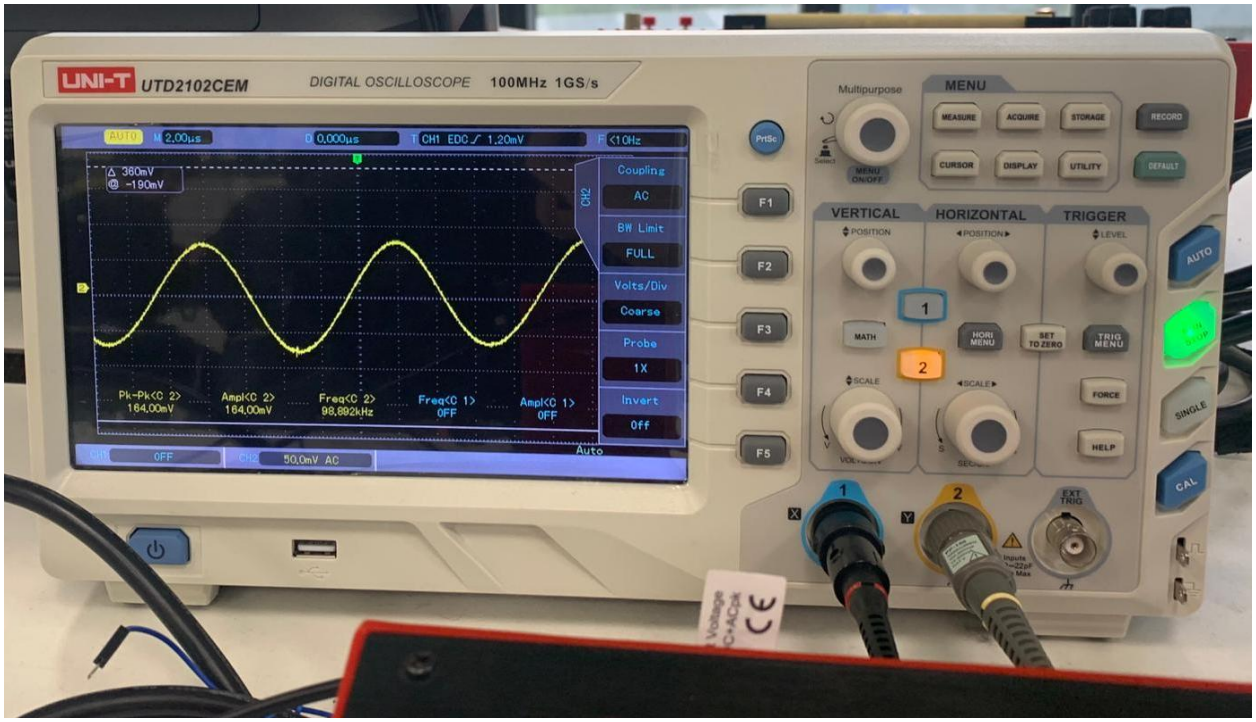


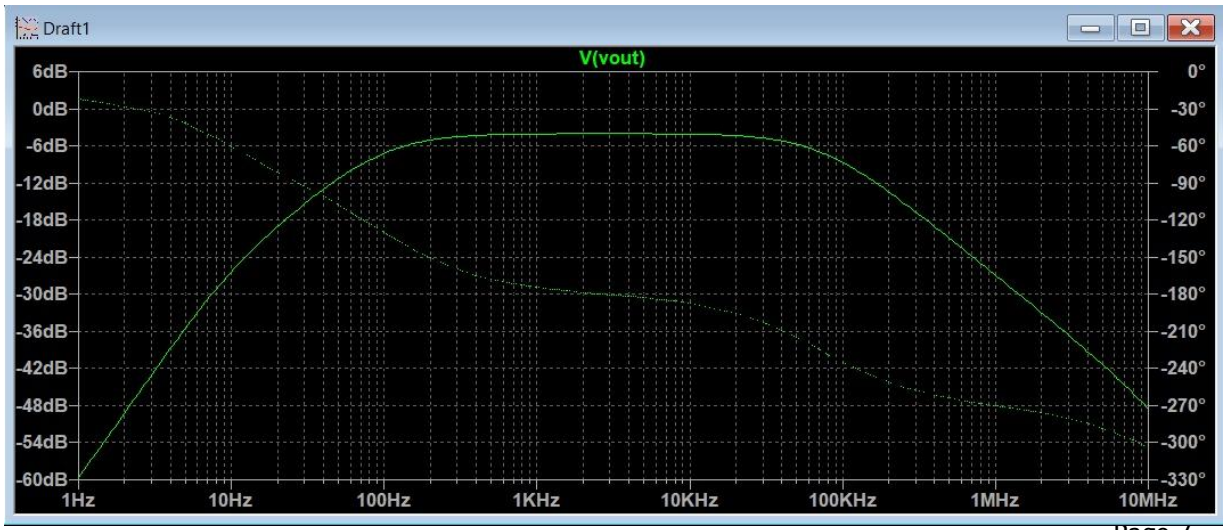
10k Hz




Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-300°
-330°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

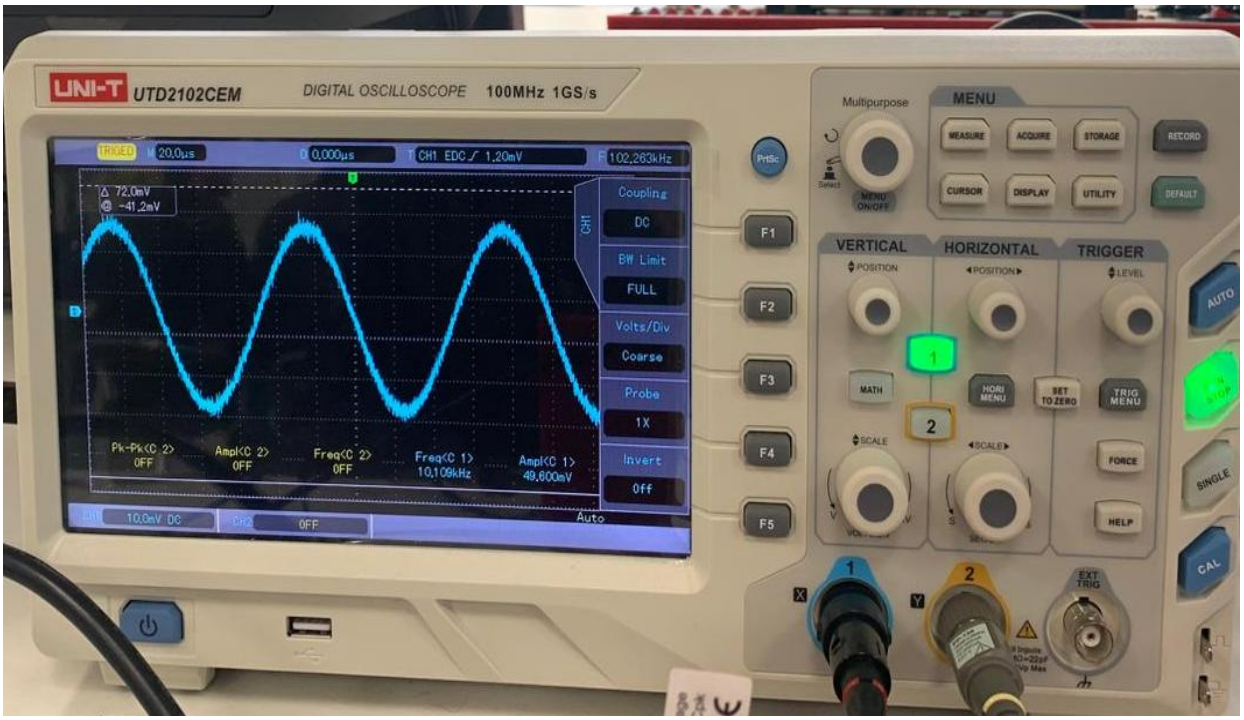
100k Hz

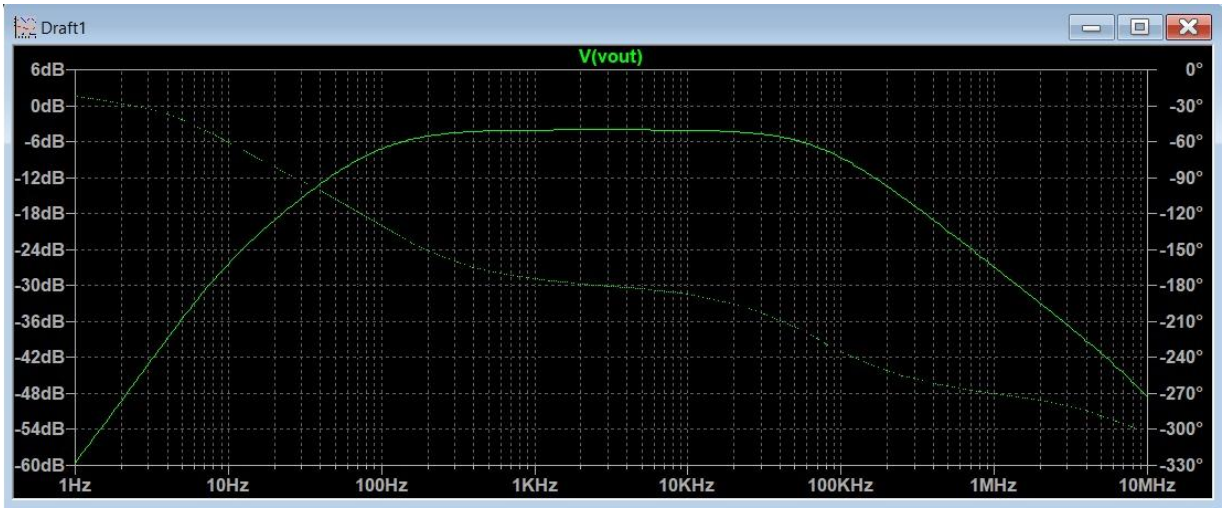

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DIGITAL OSCILLOSCOPE 100MHz 1GS/s
Multipurpose
MENU
MEASURE
ACQUIRE
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RECORD
CURSOR
DISPLAY
UTILITY
DEFAULT
VERTICAL
HORIZONTAL
TRIGGER
Coupling
AC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
F1
F2
F3
F4
F5
AUTO
SINGLE
CAL
FORCE
HELP
MATH
SET TO ZERO
EXT TRIG


Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-300°
-330°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

Freq. Response of Amplifiers Experiment 3

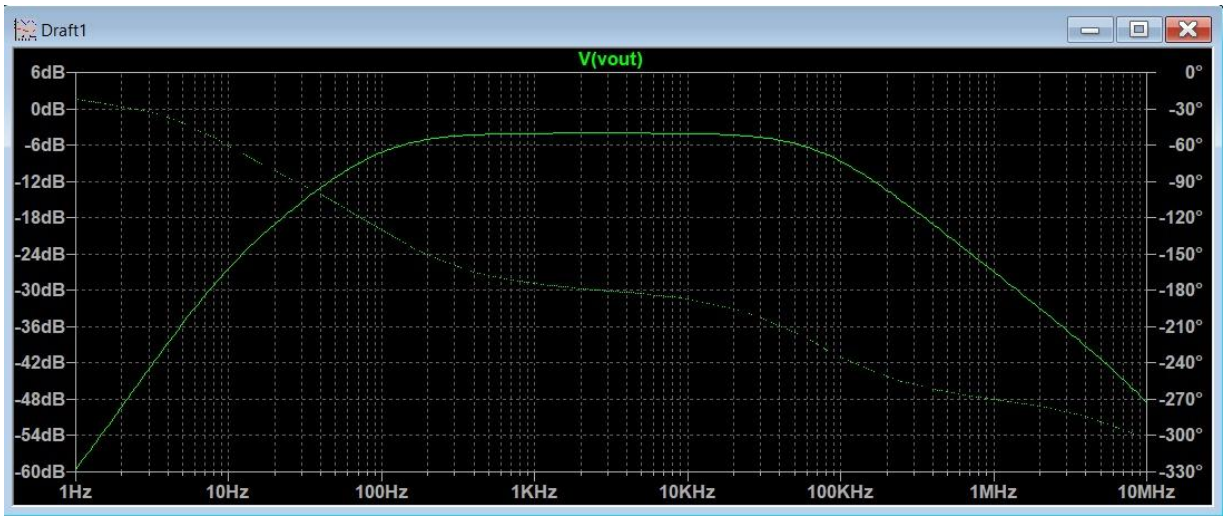
1M Hz

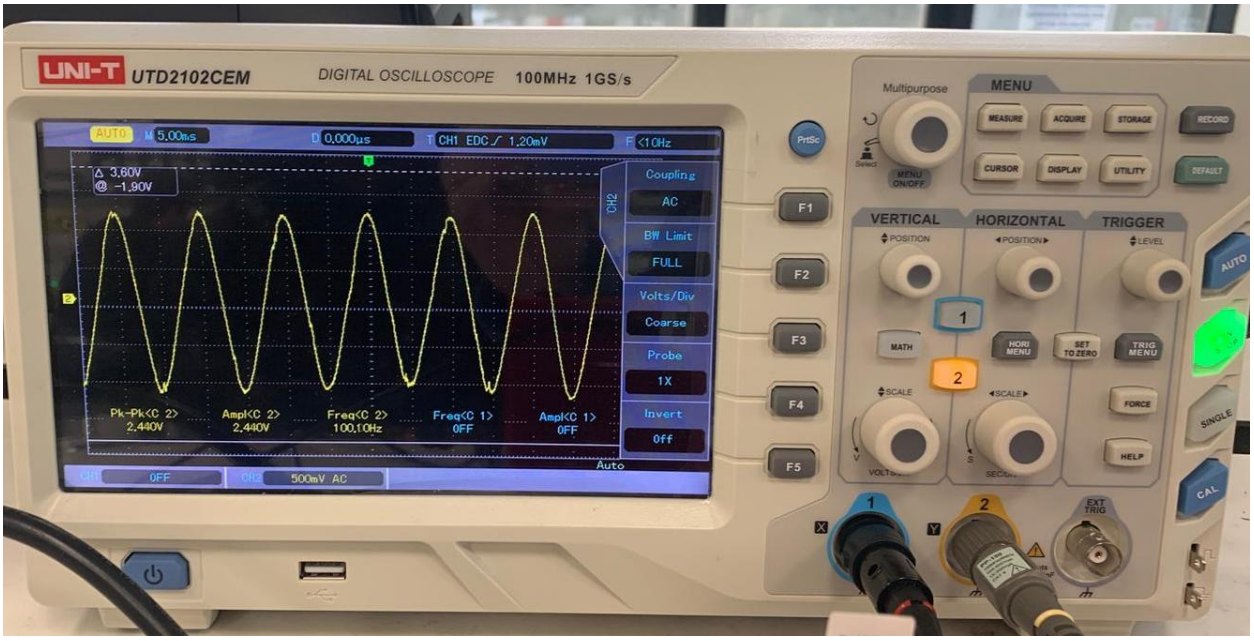


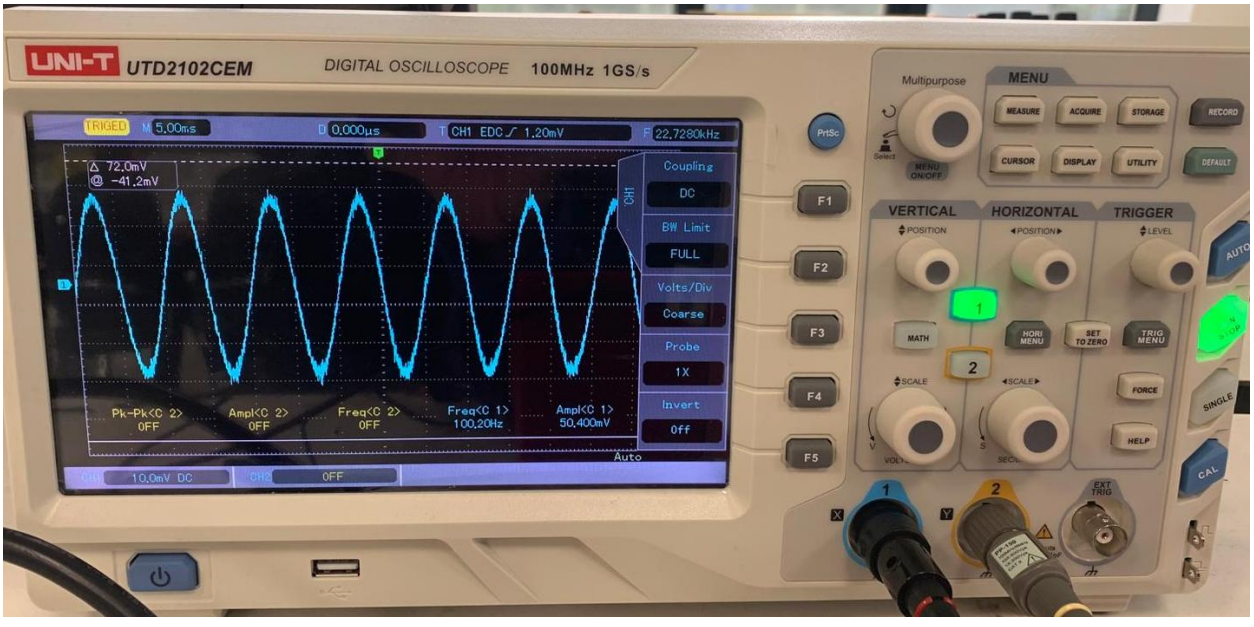


10M Hz



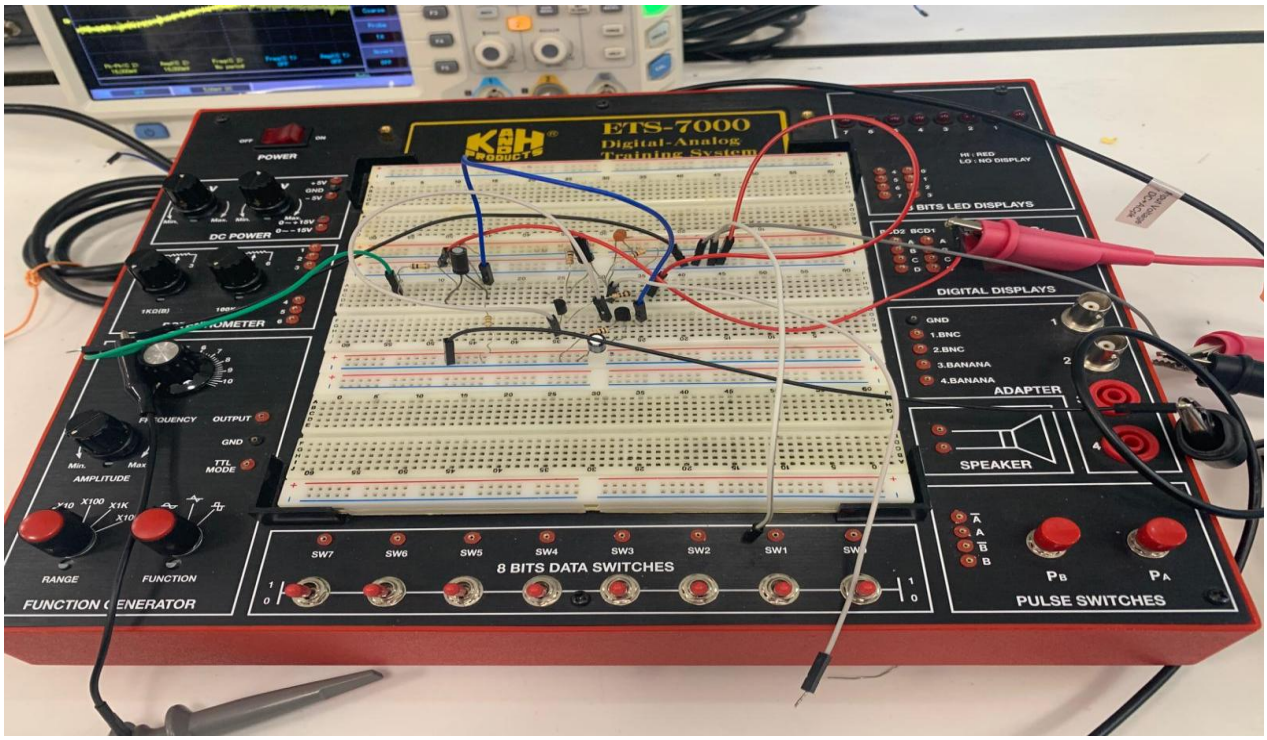

Draft1
V(vout)
6dB
0dB
-6dB
-12dB
-18dB
-24dB
-30dB
-36dB
-42dB
-48dB
-54dB
-60dB
0°
-30°
-60°
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-300°
-330°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz


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100MHz 1GS/s
Coupling
AC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
MENU
VERTICAL
HORIZONTAL
TRIGGER

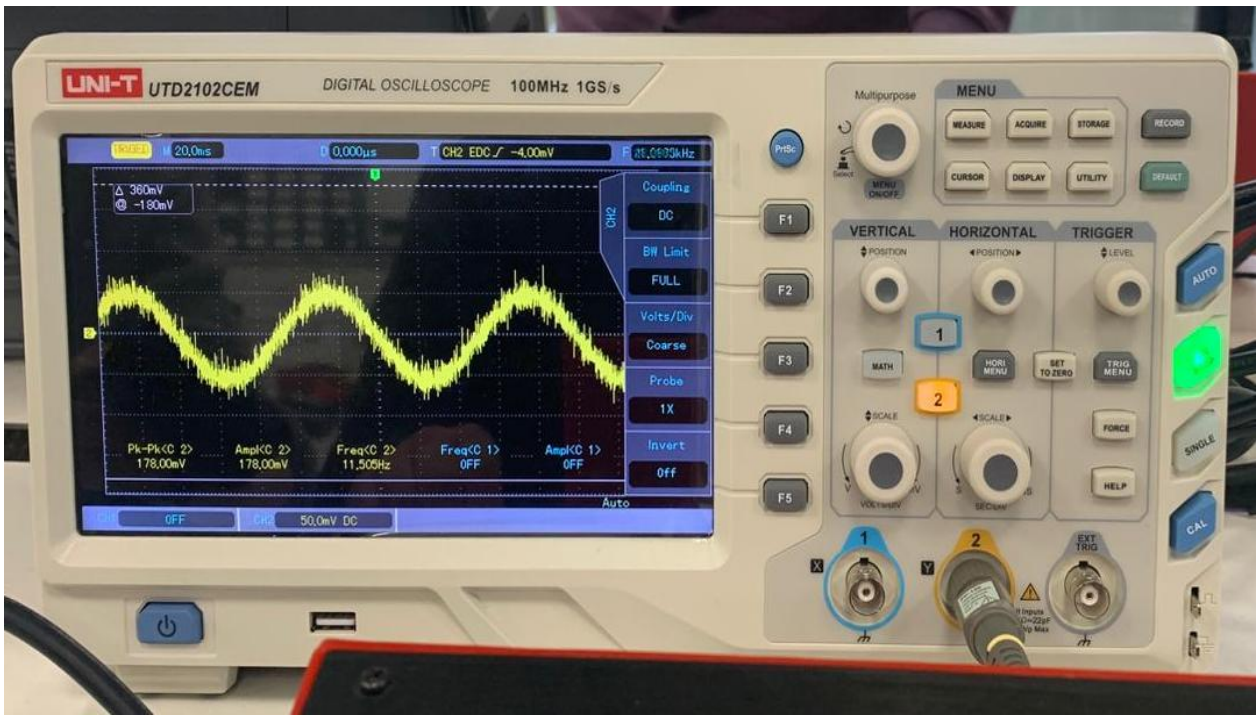

UNI-T
UTD2102CEM
DIGITAL OSCILLOSCOPE
100MHz 1GS/s
Coupling
DC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
MENU
VERTICAL
HORIZONTAL
TRIGGER

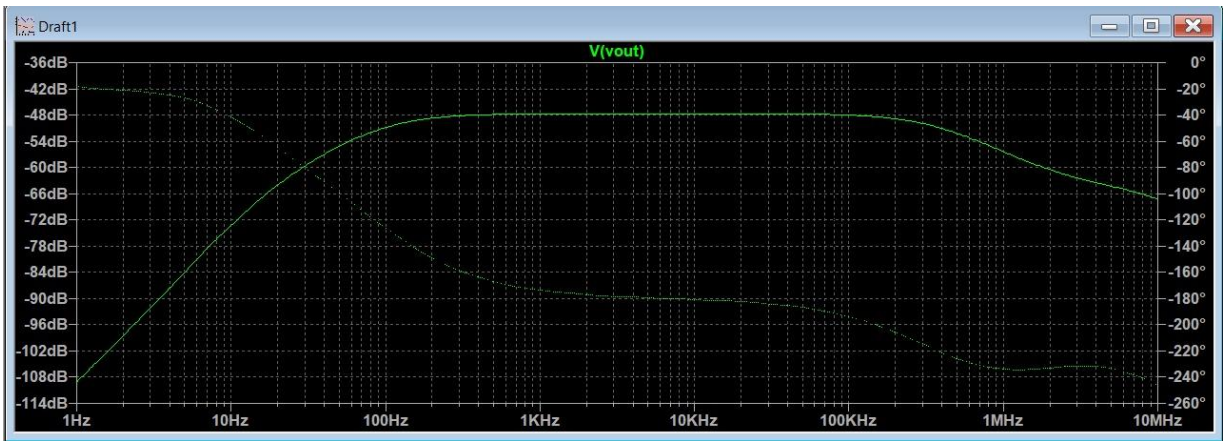
12. Power down and disassemble the circuit.
13. Set-up the circuit in Fig. 2. Be very careful while connecting the capacitor. The +/- pins must be connected as shown in the Fig. 2.
14. Connect the ground and then Vee and Vcc. Power up your circuit.
15. Apply sine waves of 10Hz, 100Hz, 1kHz, 10kHz, 100kHz, 1MHz, 10MHz with amplitudes of 50mV to Vin.
16. Also find the 3-dB cut-off frequencies.
17. Plot the gain in dB as a function of the frequency. Indicate the low (f_L) and high (f_H) cut-off frequencies. **Hard to find cut-off frequencies because of derivation**

2.Devre

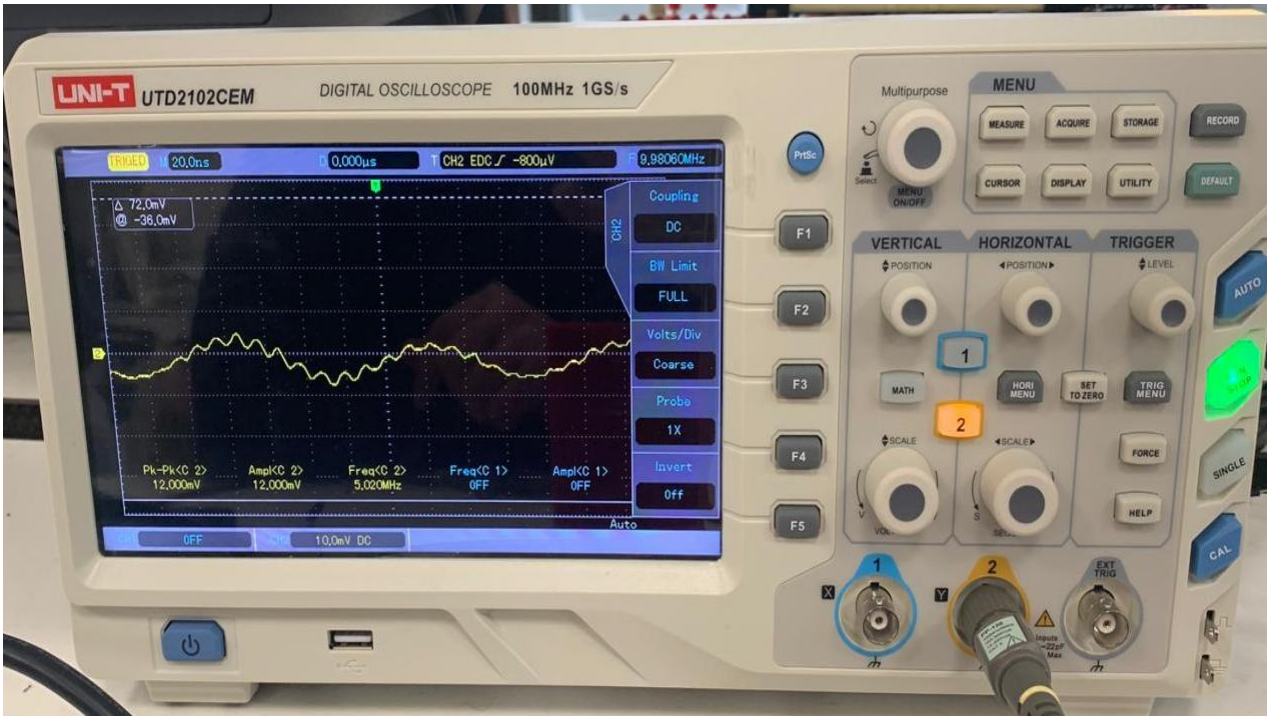


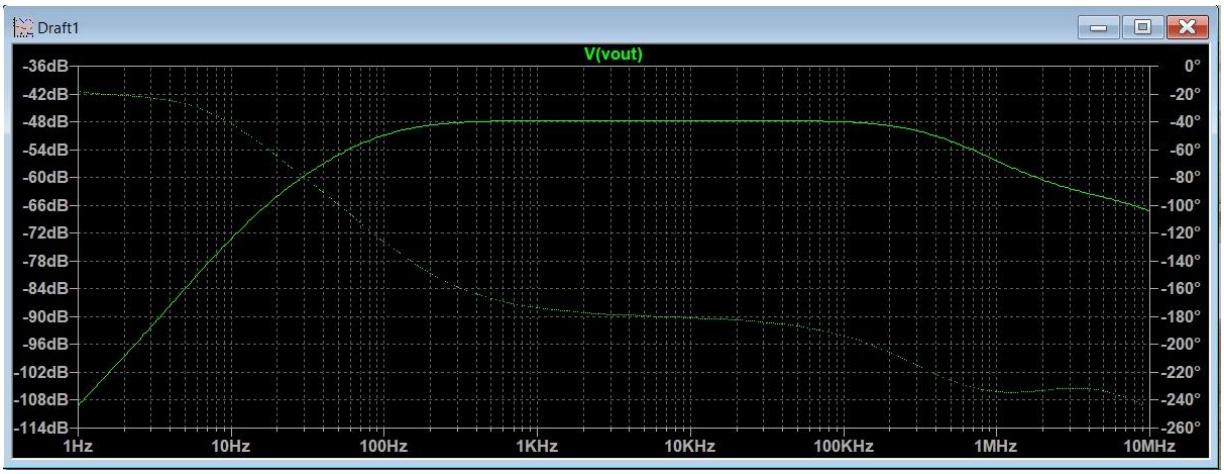
10Hz




Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
-200°
-220°
-240°
-260°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

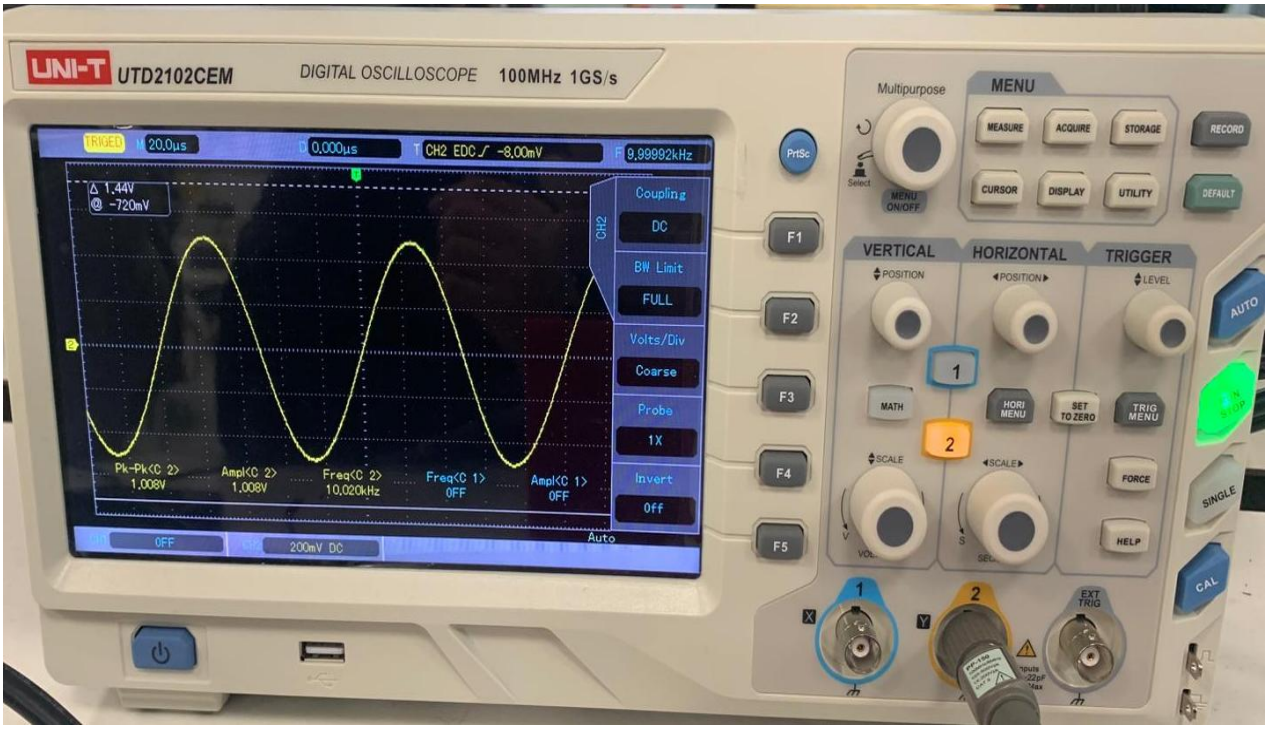
100 Hz

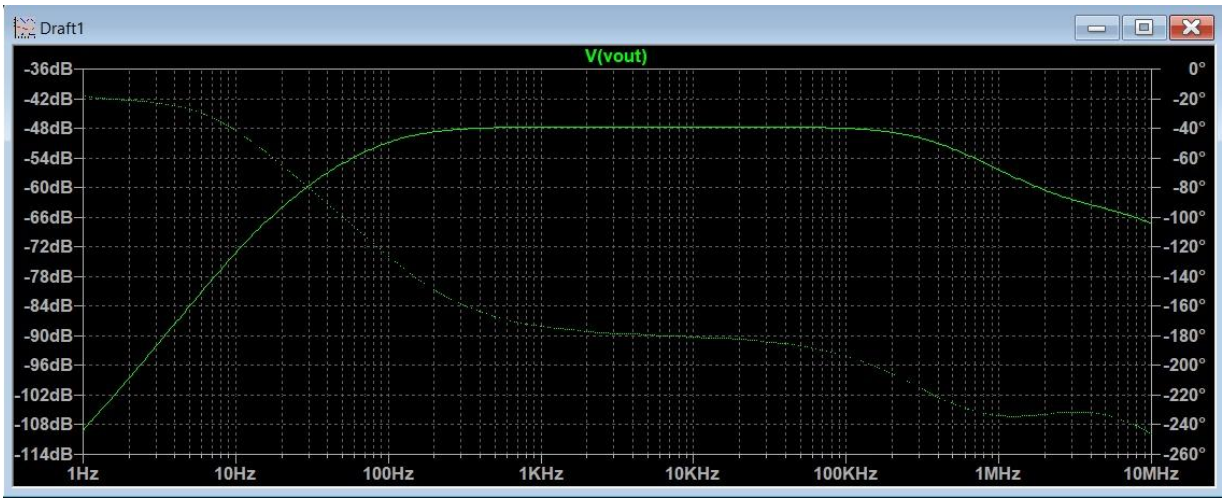

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RECORD
CURSOR
DISPLAY
UTILITY
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VERTICAL
HORIZONTAL
TRIGGER
AUTO
SINGLE
CAL
HELP
FORCE
Coupling
DC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
F1
F2
F3
F4
F5


Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
-200°
-220°
-240°
-260°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

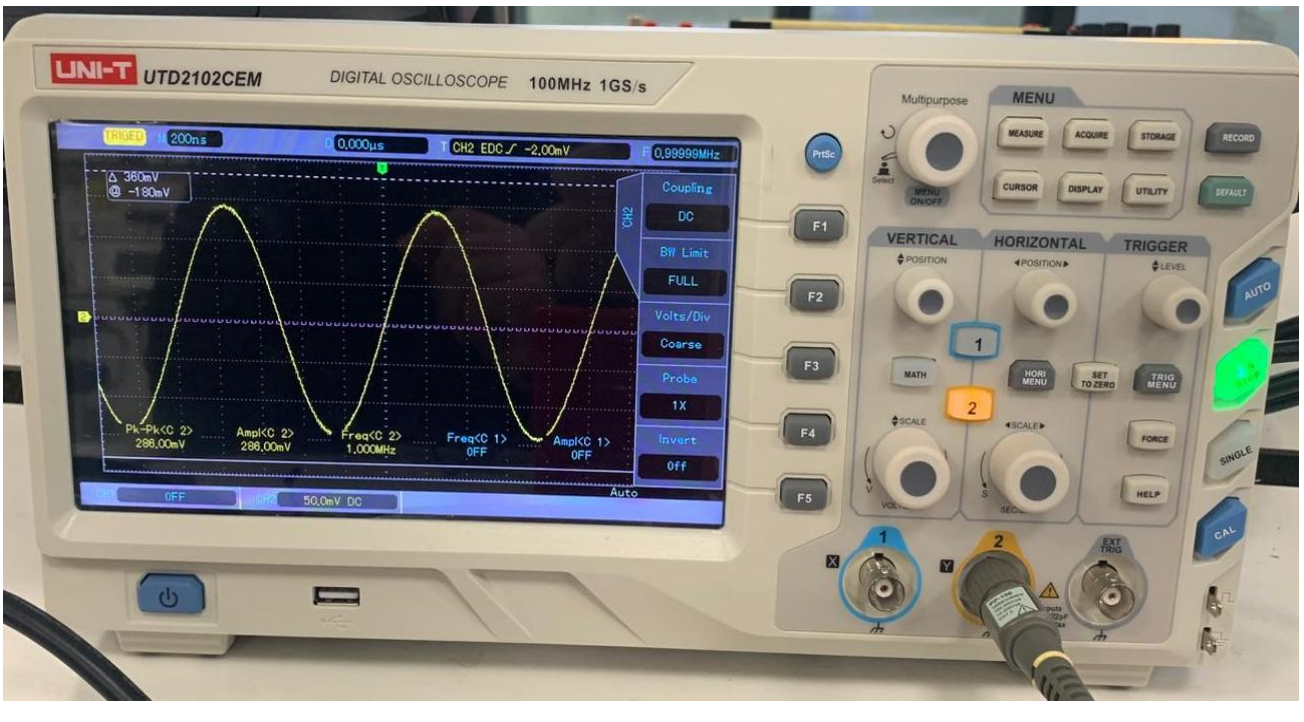
Freq. Response of Amplifiers Experiment 3

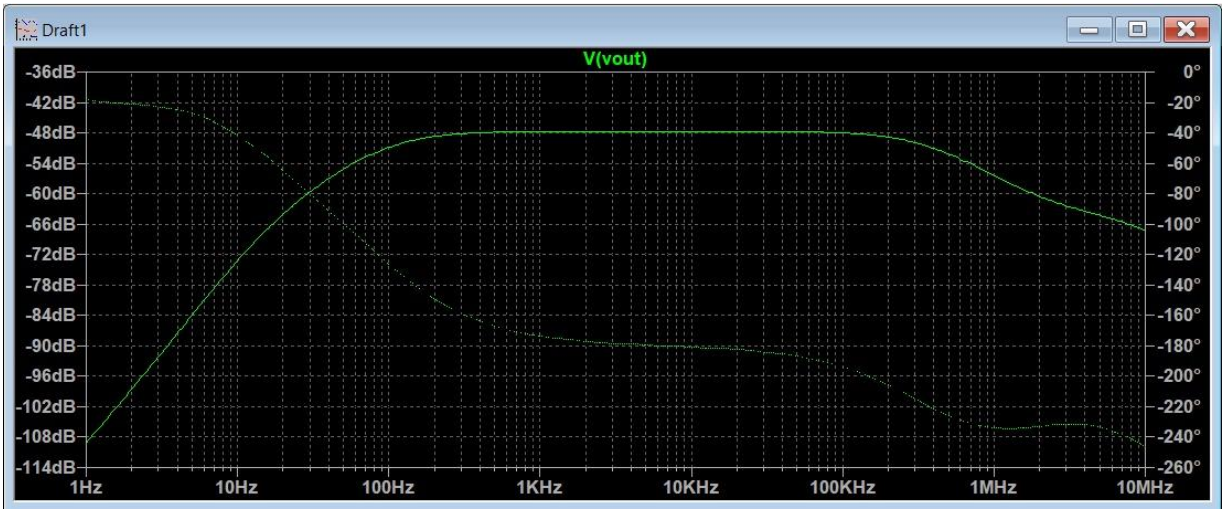
1k Hz



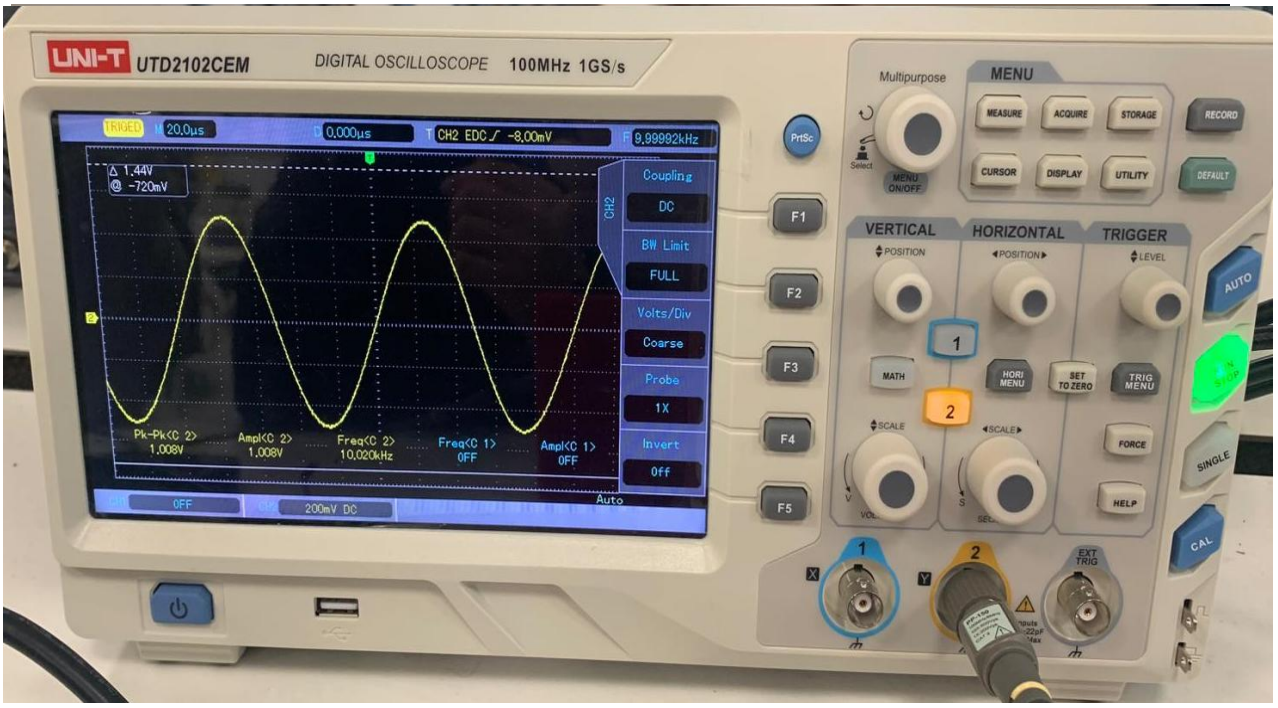


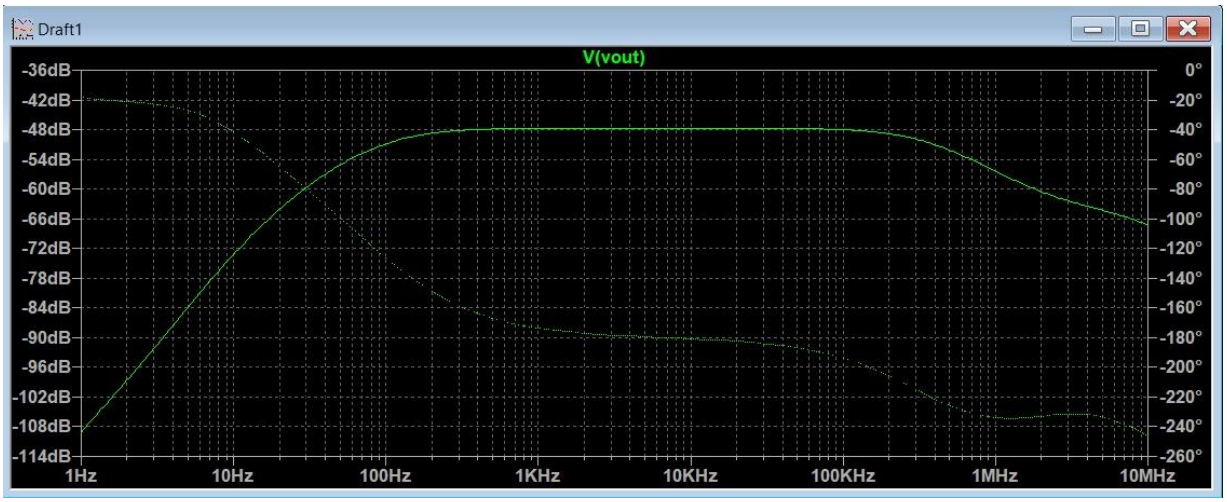
10k Hz



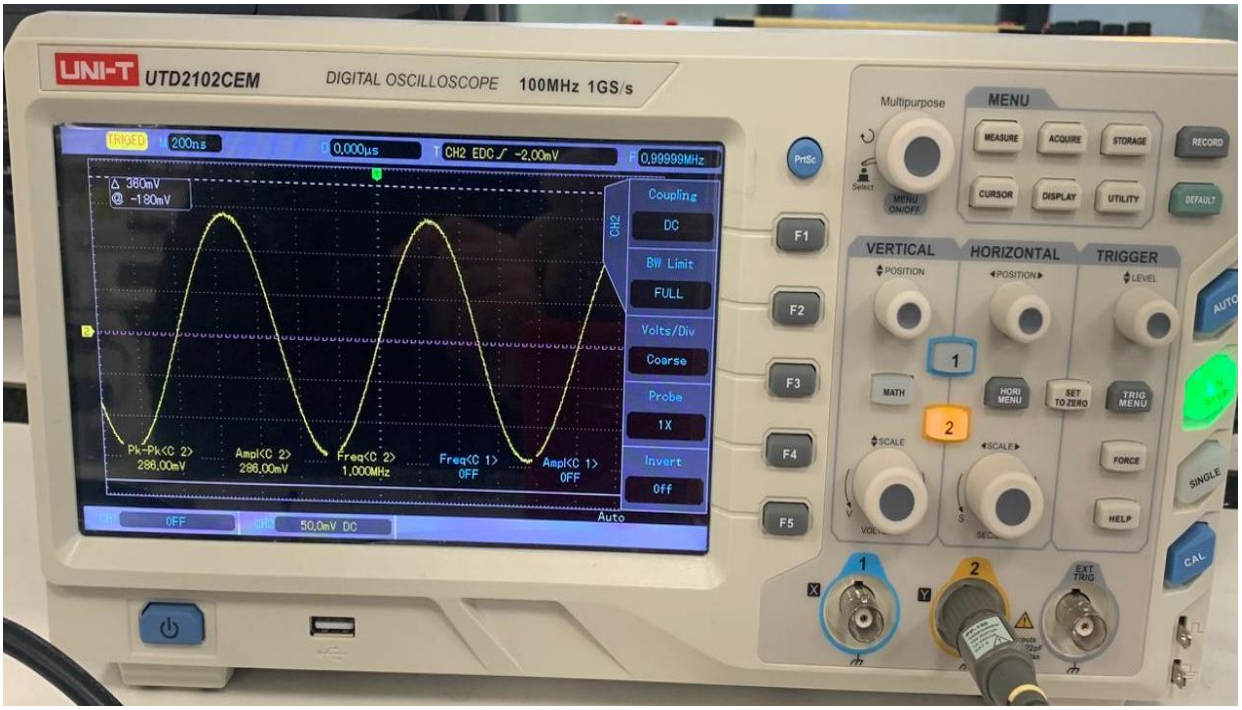

Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
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10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

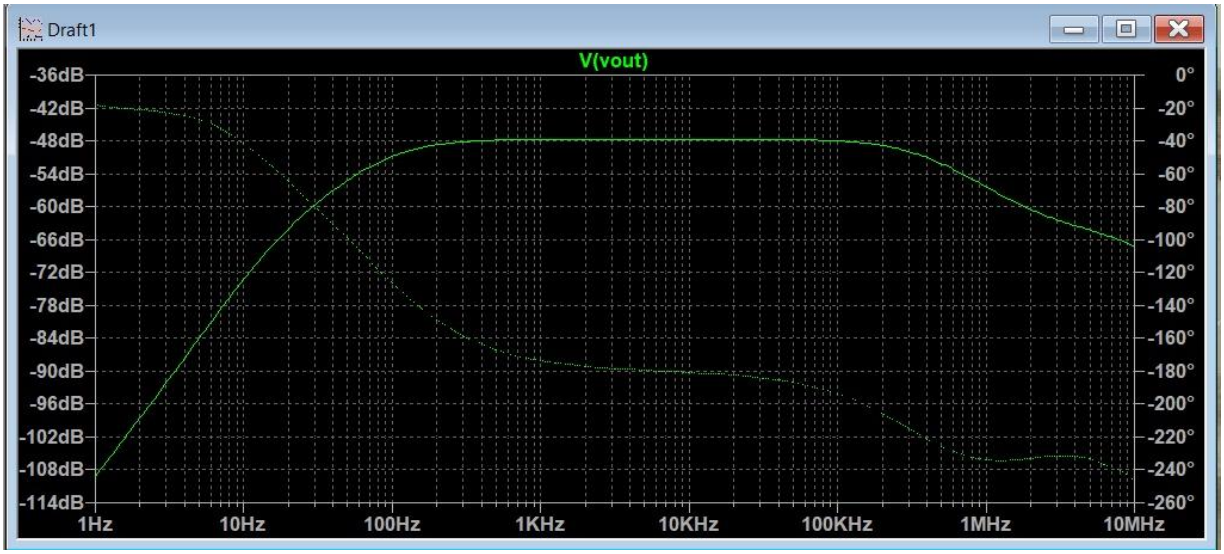
100k Hz


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DEFAULT
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MATH
HORI MENU
SET TO ZERO
TRIG MENU
FORCE
SINGLE
HELP
CAL
EXT TRIG
F1
F2
F3
F4
F5
Coupling
DC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
Pk-Pk<C 2>
1.008V
Ampl<C 2>
1.008V
Freq<C 2>
10.020kHz
Freq<C 1>
OFF
Ampl<C 1>
OFF

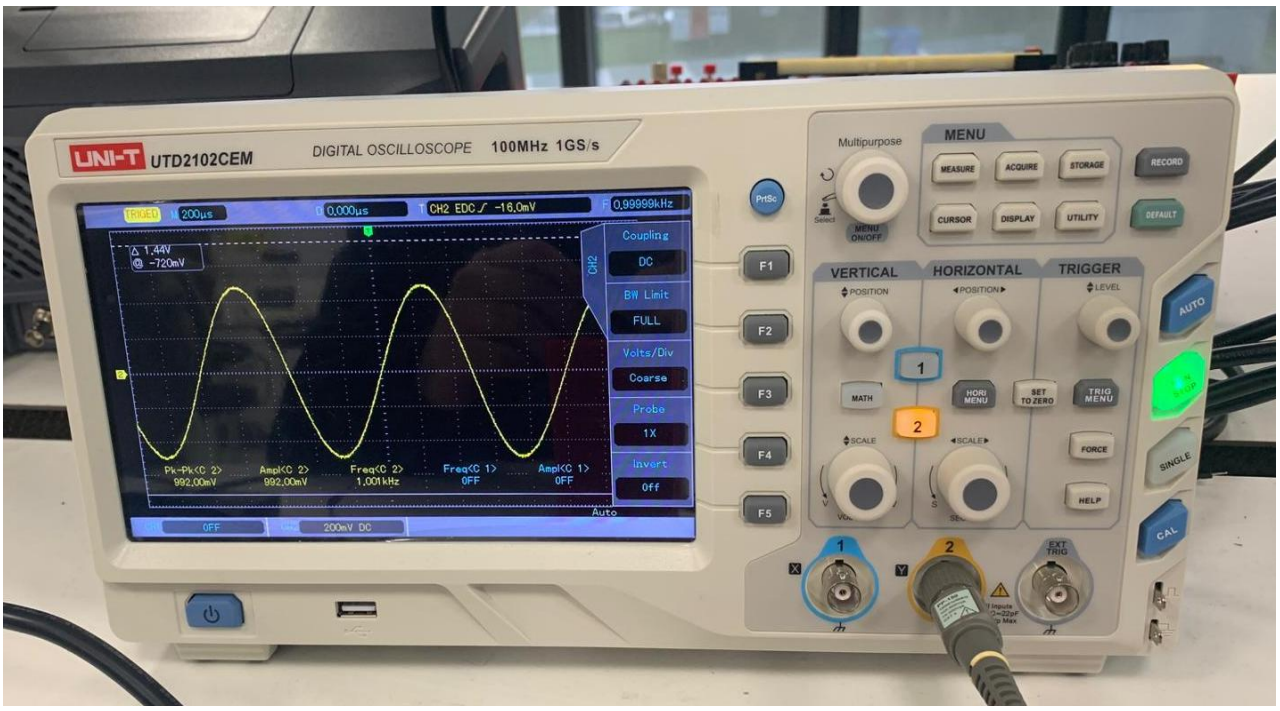

Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
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1KHz
10KHz
100KHz
1MHz
10MHz

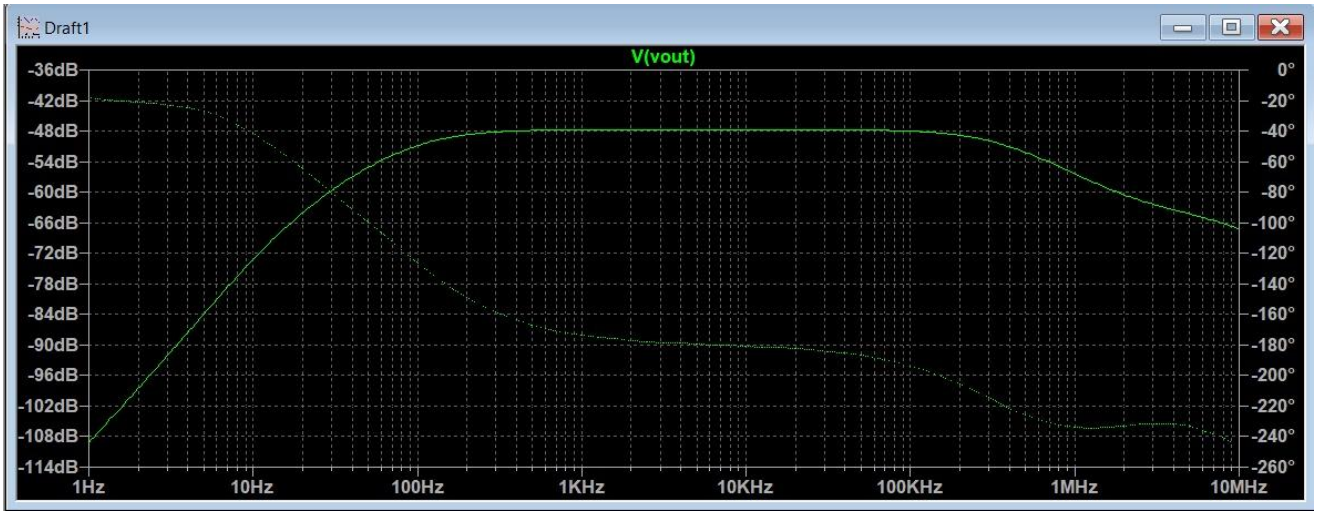
1M Hz


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MEASURE
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RECORD
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DISPLAY
UTILITY
DEFAULT
VERTICAL
HORIZONTAL
TRIGGER
MATH
HORI MENU
SET TO ZERO
TRIG MENU
FORCE
SINGLE
HELP
CAL
F1
F2
F3
F4
F5
Coupling
DC
BW Limit
FULL
Volts/Div
Coarse
Probe
1X
Invert
Off
Auto
50.0mV DC


Draft1
V(vout)
-36dB
-42dB
-48dB
-54dB
-60dB
-66dB
-72dB
-78dB
-84dB
-90dB
-96dB
-102dB
-108dB
-114dB
0°
-20°
-40°
-60°
-80°
-100°
-120°
-140°
-160°
-180°
-200°
-220°
-240°
-260°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

10M Hz





19. Compare the calculated (step 1, 3), simulated (step 2, 4) and measured (step 10, 16) responses.

Answer:

Our values which we measure on ltspice and the values was measure on lab are close but our measured values in lab are a little bit different from others. We built our both two circuits correctly and measure the values and We didnt think we have a trouble with ltspice. Because of our calculations and simulation measurements are close, we think maybe we had a mistake when we were measuring in labratory. Maybe it is about the cables, breadbords or ac source.